Tuza's Conjecture for Graphs with at Most Four Edge-Disjoint Triangles

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Abstract

For a finite simple graph G, let $\nu(G)$ be the maximum number of pairwise edge-disjoint triangles and let $\tau(G)$ be the minimum number of edges meeting every triangle. We prove that $\tau(G) \leq 2\nu(G)$ whenever $\nu(G) \leq 4$. The cases $\nu(G) \leq 3$ were proved by Parker. For $\nu(G) = 4$, a discharging reduction based on the method of Sheng and Xiao shows that a counterexample would have triangle support on at most twelve vertices. Marking a maximum four-packing leaves 66 finite order-type cases. We encode these cases as SAT instances and certify each resulting formula by an LRAT refutation. A Lean formalization reconstructs the semantic clauses, checks the exact formulas and all 66 refutations, and derives the stated graph theorem.

1 Introduction

A triangle packing in a graph is a family of pairwise edge-disjoint triangles, and a triangle transversal is a set of graph edges meeting every triangle. Tuza conjectured that every finite simple graph G satisfies

$$\tau(G) \leq 2\nu(G). \tag{1}$$

The factor two is best possible already for K_4 and K_5. The conjecture remains open in general; see Tuza's original paper [9] and the subsequent literature cited by Parker [5].

The best universal quantitative bound currently available is weaker than factor two; Haxell's bound [3], for example, gives only $\tau(G) \leq 11$ when $\nu(G) = 4$. Many exact results instead impose a graph class, such as maximum average degree below seven [6] or maximum degree at most seven [2]. Parker proved the exact bounded-packing statement

$$\nu(G) \leq 3 \quad \Longrightarrow \quad \tau(G) \leq 2\nu(G). \tag{2}$$

We establish the next bounded-packing case.

Theorem 1.1. *Every finite simple graph G with $\nu(G) \leq 4$ satisfies $\tau(G) \leq 2\nu(G)$.*

The proof has an analytic part and a computer-assisted finite part. The analytic part adapts the $3k$ discharging framework of Sheng and Xiao [7]. Starting from a maximal packing saturated under three local exchanges, the argument either finds an exact fat-head crown or bounds the triangle support by three times the packing size. Exact offset identities for crowns and triangle-isolated copies of K_4 then give the following reduction.

Proposition 1.2. *If $\nu(G) = 4$ and $\tau(G) \geq 9$, then the triangle support of G has at most twelve vertices.*

For the finite part, mark four edge-disjoint triangles. Their vertex intersection pattern is a linear 3-uniform hypergraph with four edges. There are 66 order-type scopes on orders six through twelve. In each scope, all other graph edges remain free. A SAT formula represents the literal graph and its complete triangle set. Whenever a model has five edge-disjoint triangles or a transversal of at most eight edges, a semantic clause recording that exact witness is added. Exhaustion yields 66 final formulas. Their DRAT proofs were converted to LRAT and checked in Lean; the original DRAT traces were also checked independently with `drat-trim` [4].

The finite verification does not rely on a catalogue of unlabeled graphs or on the solver's UNSAT declaration. The reduction to twelve vertices, completeness of the marked support types, graph-to-CNF translation, and case dispatch are part of the Lean proof.

2 Preliminaries

All graphs are finite, undirected, loopless, and simple. Write $\mathcal{T}(G)$ for the set of triangles of G. We identify a triangle with either its three vertices or its three graph edges when no ambiguity is possible.

The *triangle packing number* $\nu(G)$ is the largest cardinality of a subfamily of $\mathcal{T}(G)$ whose members are pairwise edge-disjoint. The *triangle transversal number* $\tau(G)$ is the minimum size of a set $C \subseteq E(G)$ such that C meets the edge set of every triangle. Define the integral Tuza defect

$$\operatorname{def}(G) = \tau(G) - 2\nu(G).$$

A strict counterexample to (1) is therefore a graph with $\operatorname{def}(G) > 0$.

The *triangle support* $\operatorname{supp}_{\triangle}(G)$ is the graph whose edge set consists of the edges lying in at least one triangle of G, and whose vertex set consists of their endpoints. Then

$$\nu(\operatorname{supp}_{\triangle}(G)) = \nu(G), \qquad \tau(\operatorname{supp}_{\triangle}(G)) = \tau(G). \tag{3}$$

For a vertex v, form the auxiliary graph whose vertices are the graph edges incident with v, with two such edges adjacent when they lie together in a triangle. If this auxiliary graph is disconnected, split v into one vertex for each component, attaching each incident edge to the corresponding copy. The edge ground set is unchanged, and three edges form a triangle before the split exactly when their corresponding edges form a triangle afterward. Consequently a sequence of such splits preserves the entire triangle hypergraph, and hence preserves both ν and τ. We call the resulting graph *split-normal*.

We need one further reduction object from Sheng and Xiao.

Definition 2.1 (fat-head crown)**.** A fat-head crown in a graph $G = (V, E)$ is a triple (C, H, X) such that C and X partition V, $H \subseteq E$, and:

(i) C is independent;

(ii) H is exactly the set of edges spanned by vertices of C, meaning $uv \in H$ exactly when cuv is a triangle for some $c \in C$;

(iii) no vertex of C is adjacent to a vertex of $X \setminus V(H)$; and

(iv) there is an edge-disjoint packing P of $|H|$ triangles, each with exactly one vertex in C and exactly one edge in H.

The packing P is a *witness packing*.

The exact witness requirement is essential. A subset passing a relaxed matching test need not be a crown.

Lemma 2.2 (exact crown offsets)**.** *Let (C, H, X) be a fat-head crown with witness packing P, put $h = |H|$, and let $G' = G - C - H$. Then*

$$\nu(G) = \nu(G') + h, \qquad \tau(G) = \tau(G') + h.$$

Proof. Adjoining P lifts every packing of G'. Conversely, delete from a packing of G every triangle meeting C or H. Each deleted triangle contains a head edge, and distinct packed triangles contain distinct head edges, so at most h triangles are deleted.

Adjoining H lifts every transversal of G'. Conversely, a transversal of G meets the h pairwise edge-disjoint triangles of P in at least h distinct graph edges. These edges disappear on intersection with $E(G')$, and that intersection remains a transversal of G'. Both identities follow. □

Lemma 2.3 (triangle-isolated K_4 edge block)**.** *Suppose a copy $Q \cong K_4$ has no edge in a triangle outside the four faces of Q, and let $G' = G - E(Q)$. Then*

$$\nu(G) = \nu(G') + 1, \qquad \tau(G) = \tau(G') + 2.$$

Proof. At most one face of Q can occur in an edge-disjoint packing, while any one face can be added to a packing of G'. A transversal of the four faces uses at least two local edges, and any pair of opposite edges meets all four faces. The isolation hypothesis separates these local choices from G'. □

3 The twelve-vertex reduction

We isolate the form of the Sheng–Xiao discharging argument needed here. Let S be a maximal triangle packing, let $F = V(G) \setminus V(S)$, and let $R = E(G) \setminus E(S)$. We allow the following packing-only improvements:

(E1) replace one member of S by at least two triangles using only R and the three edges of the removed triangle;

(E2) replace two members of S by at least three triangles using only R and the six edges of the removed triangles;

(E3) replace two members T_1, T_2 by two triangles T_1', T_2' such that $(S \setminus \{T_1, T_2\}) \cup \{T_1', T_2'\}$ is a packing whose vertex support is strictly larger than the vertex support of S.

After each exchange, extend greedily to a maximal packing. The lexicographic pair $(|S|, |V(S)|)$ strictly increases and is bounded by $(\nu(G), 3\nu(G))$, so the process terminates.

An edge $uv \in E(S)$ is *labelled* if some $x \in F$ spans it, that is, xuv is a triangle. Let L be the set of labelled edges.

Lemma 3.1 (discharge bound)**.** *Let G be triangle-supported and split-normal. Assume that:*

(a) G has no triangle-isolated K_4 edge block as in Lemma 2.3;

(b) S is a maximal packing saturated under (E1)–(E3); and

(c) there is no nonempty fat-head crown (C, H, X) with $C \subseteq V(G) \setminus V(L)$ and $H \subseteq L$.

Then $|V(G)| \leq 3|S|$.

Proof. We give the charge argument in full.

First, each triangle of S has at most one labelled edge. If two of its edges are spanned by distinct vertices of F, (E1) applies. Suppose instead that $x \in F$ spans uv and vw in the packed triangle uvw. The vertices u, v, w, x induce K_4. Since it is not a triangle-isolated edge block, some edge belongs to an external triangle. Up to interchanging u and w, there are four edge orbits, and all four give a permitted exchange:

(i) if the labelled edge vw lies in vwy, use xuv and vwy for (E1) when neither vy nor wy lies in $E(S)$. If vy lies in a retained packing triangle, retain that triangle and insert xuw; if wy lies in a retained packing triangle, retain it and insert xuv. Either choice is a safe (E3) exchange, even when both edges belong to retained packing triangles;

(ii) if the outside spoke xw lies in xwy, maximality forces wy into a packing triangle; retain that triangle and insert xuv by (E3);

(iii) if the opposite edge uw lies in uwy, use xuv and uwy for (E1), unless uy or wy belongs to $E(S)$; in those cases retain its packing triangle and insert xvw or xuv, respectively;

(iv) if the central spoke xv lies in xvy, maximality forces vy into a packing triangle; retain it and insert xuw.

Here an external triangle means a triangle other than the four faces of the displayed K_4. In every two-for-two move, the inserted face contains one edge of a removed packing triangle and two spokes incident with $x \in F$. It is therefore edge-disjoint from every retained packing triangle. Moreover the retained triangle keeps every old vertex omitted by the inserted face, while the new face adds $x \notin V(S)$. Each two-for-two move is consequently a globally safe (E3) exchange with strictly larger vertex union. Thus at most one edge of each member of S is labelled.

Call a packing triangle *bad* if it has no labelled edge and *good* otherwise. Delete L temporarily. A good triangle is *excellent* if neither unlabelled edge lies in another remaining triangle, and *pretty-good* otherwise. Write S_1, S_2, S_3 for the excellent, pretty-good, and bad triangles. Put

$$V_1 = V(S_1) \setminus \big(V(L) \cup V(S_2) \cup V(S_3)\big), \qquad V_2 = V(S_2) \setminus V(L).$$

The third vertex of every good packing triangle lies outside $V(L)$. Indeed, write such a triangle as uvw, with labelled edge uv spanned by $q \in F$, and suppose that $w \in V(L)$. A different retained packing triangle then contains a labelled edge incident with w. Retain that triangle and replace uvw by quv. The result is a packing: uv belonged to the removed triangle, while the two edges incident with $q \notin V(S)$ belong to no member of S. The retained triangle preserves w, the new triangle preserves u, v, and q is a new support vertex. This is an (E3) improvement, a contradiction. In particular, every component meeting S_2 contains a vertex of V_2.

Place three units on each labelled edge and on each bad triangle. Since every member of $S_1 \cup S_2$ has exactly one labelled edge, the total charge is exactly $3|S|$. Move one unit from each labelled edge to each endpoint, and move one unit from each bad triangle to each of its vertices. Each labelled edge retains one unit. At this point only vertices in $F \cup V_1 \cup V_2$ can have charge zero.

Consider a connected component of the graph formed by the edges of S. If it contains x packing triangles and covers m vertices, form the bipartite incidence graph between those triangles and those vertices. This incidence graph is connected, has $x + m$ vertices, and has exactly $3x$ edges. Hence $x + m \leq 3x + 1$, so $m \leq 2x + 1$. If x_2 of the component's triangles are bad, the charge available on

its vertices after the first move is $2x + x_2$. Move surplus charge inside the component, prioritizing vertices of V_1. Each component then has at most one zero; a component meeting S_3 has none, and a component meeting S_2 can have a zero only in V_2.

We next establish the component transfer and the residual-zero statements that complete the discharging step; compare Sheng–Xiao [7, Lemmas 8–12].

Claim 3.2 (componentwise transfer)**.** *For every packing component whose unique zero is a vertex $w \in V_2$, there are two labelled edges in that component with one common, unique spanner $q \in F$. Moving one of their remaining units to w and the other to q makes w positive. This can be done once in every such component without using any labelled edge twice. Whenever a labelled edge is reduced to zero, its unique spanner in F receives a unit in the same component transfer.*

Proof of the claim. Choose the pretty-good triangle $wuv \in S_2$ containing w, with labelled edge uv spanned by some $q \in F$. Since it is pretty-good, after possibly interchanging u and v, the unlabelled edge uw lies in a triangle uwx of $G - L$. If neither ux nor wx lies in $E(S)$, then quv and uwx replace wuv by (E1), a contradiction. If wx lies in a packing triangle, retain that triangle and replace wuv by quv. The two spokes at q avoid every retained member, and the new pair gains the vertex q, so this is a globally safe (E3), again a contradiction. We may therefore assume that ux lies in a packing triangle uxy and that $wx \notin E(S)$.

The triangle uxy is not bad, since its packing component still has a zero after the component redistribution. Its labelled edge is not ux, because uwx lies in $G - L$. If its labelled edge were xy, with spanner $q' \in F$, then retaining wuv and replacing uxy by $q'xy$ would be a globally safe (E3) exchange whose vertex union gains q'. Hence uy is labelled. The triangle uxy is pretty-good because its unlabelled edge ux lies in $uwx \subseteq G - L$.

The labelled edges uv and uy have one common unique spanner. Indeed, if $q_1 \in F$ spans uv and $q_2 \in F$ spans uy with $q_1 \neq q_2$, then the three triangles

$$q_1uv, \qquad q_2uy, \qquad uwx$$

are pairwise edge-disjoint and use only edges outside $E(S)$ together with edges of the two removed triangles wuv and uxy; recall that $wx \notin E(S)$. They give (E2). Applying this observation with every choice of spanner shows both uniqueness and equality; call the common vertex q. Transfer the unit on uv to w and the unit on uy to q.

There is at most one zero in a packing component. We make the displayed choice only once for that component. Distinct components contain disjoint packing edges, so their two labelled edges are disjoint and no charge is spent twice. The last assertion follows from the construction: every edge that loses its last unit is one of the displayed pair, and the common spanner q receives the other edge's unit in the same transfer. □

Perform the transfer of Claim 3.2 in every eligible component. Every vertex in V_2 is now positive. Let I be the zero-charge vertices remaining in $F \cup V_1$, and let L' be the labelled edges that still carry one unit.

Claim 3.3 (residual independence)**.** *The set $F \cup V_1$, and hence I, is independent.*

Proof of the claim. Suppose that uv is an edge with $u, v \in F \cup V_1$. Triangle support gives a triangle uvw, and maximality of S forces at least one of its three edges into $E(S)$. If $uv \in E(S)$, then $u, v \in V_1$ and the packing triangle containing uv is neither bad nor good: it cannot be bad because V_1 avoids $V(S_3)$, while each possible labelled edge of a good triangle has endpoint u or v, contrary to $V_1 \cap V(L) = \varnothing$.

Otherwise, say $uw \in E(S)$. Then $u \in V_1$, and the packing triangle containing uw belongs to S_1, since u lies in no triangle of $S_2 \cup S_3$. None of uv, vw, uw is labelled because neither u nor v lies in $V(L)$. Thus the unlabelled packing edge uw lies in the other triangle uvw of $G - L$, contradicting excellence. Both cases are impossible. □

Claim 3.4 (outside residual zeros)**.** *Every vertex of $I \cap F$ spans only edges of L'.*

Proof of the claim. A vertex of F can span only an edge of $E(S)$: otherwise its triangle would be edge-disjoint from the maximal packing. Such an edge is labelled. If $z \in I \cap F$ spanned an edge of $L \setminus L'$, that edge must have lost its last unit in Claim 3.2, the only operation that depletes a labelled edge. The claim then sends the unit to the edge's unique spanner z, contradicting that z still has charge zero. □

Claim 3.5 (internal residual zeros)**.** *Every vertex of $I \cap V_1$ spans only edges of L'.*

Proof of the claim. Fix $v' \in I \cap V_1$ and let Y be its packing component. We first show that every neighbour of v' lies in $V(Y)$. Partition the incident edges at v' into those whose other endpoint lies in $V(Y)$ and those whose other endpoint does not. The first part is nonempty. No triangle can use one edge from each part. For if $x'v'u'$ were such a triangle and $x'v' \notin E(S)$, then none of its three edges could belong to $E(S)$—any such packing edge would join the two packing components—contradicting maximality. If $x'v' \in E(S)$, write $x'v'y' \in S$. The component Y contains only excellent triangles: a zero in a component meeting S_3 is impossible, and a zero in a component meeting S_2 lies in V_2, whereas $v' \in V_1$. Thus $x'v'y' \in S_1$, and its labelled edge is $x'y'$ because $v' \notin V(L)$. Let $q' \in F$ span $x'y'$. If $q' = u'$, then q' also spans $x'v'$, giving two labelled edges in $x'v'y'$. If $q' \neq u'$, then $q'x'y'$ and $u'x'v'$ replace $x'v'y'$ by (E1). Both alternatives are impossible. Split-normality now forces the second part of the incident star to be empty, proving $N(v') \subseteq V(Y)$.

Suppose that v' spans an edge ab. Then $a, b \in V(Y)$. If $ab \in L$, it is the labelled edge of an excellent packing triangle in Y and was not used by Claim 3.2, which uses only edges of S_2; hence $ab \in L'$. Assume instead that $ab \notin L$. If $ab \in E(S)$, write $abd \in S_1$. When $d = v'$, this packing triangle has no labelled edge, contrary to its membership in S_1. When $d \neq v'$, the triangle abv' is in $G - L$, so the unlabelled edge ab of the excellent triangle abd lies in another triangle of $G - L$, again a contradiction.

It remains that $ab \notin E(S) \cup L$. Maximality applied to the triangle abv' forces one of av' and bv' into $E(S)$; say $bv' \in E(S)$ and write $bv'c \in S_1$. Here $c \neq a$. The triangle abv' lies in $G - L$, so the unlabelled edge bv' of the excellent triangle $bv'c$ again lies in another triangle of $G - L$. This final contradiction proves the claim. □

We finish with the crown step, spelling out the matching equality. Form the bipartite graph B with left part I, right part

$$S_G(I) = \{e \in L' : e \text{ is spanned by a vertex of } I\},$$

and incidence $ce \in E(B)$ exactly when c spans e. If $|I| > |L'|$, then a maximum matching of B leaves a left vertex unmatched. Start alternating paths at all unmatched left vertices, traverse unmatched edges left-to-right and matched edges right-to-left, and let C and H be the reached left and right vertices. Every neighbour of a reached left vertex is reached; every reached right vertex is matched, or the path would augment; and its matched mate is reached. Consequently

$$N_B(C) = H = S_G(C),$$

and the matching saturates H. In particular C is nonempty.

For each matched pair ce, choose the triangle consisting of c, the edge e, and its endpoints. Notice that $C \cap V(H) = \varnothing$: vertices of F lie outside $V(S)$, vertices of V_1 lie outside $V(L)$, and every edge of $H \subseteq L$ has both endpoints in $V(L)$. The chosen triangles are therefore edge-disjoint: their head edges and crown vertices are distinct, and C is independent by Claim 3.3. Finally, if $c \in C$ had a neighbour outside $C \cup V(H)$, triangle support of that incident edge would give an edge spanned by c outside $H = S_G(C)$, a contradiction. Thus $(C, H, V(G) \setminus C)$ is an actual fat-head crown, contrary to (c).

Therefore $|I| \leq |L'|$. Choose an injection from I to L' and move the unit on each selected edge to its assigned vertex; unused edge charge may remain. Every graph vertex now has at least one unit, while total charge remained $3|S|$. Hence $|V(G)| \leq 3|S|$. □

Proof of Proposition 1.2. Assume $\nu(G) = 4$ and $\tau(G) \geq 9$. Replace G by its triangle support and split vertices until the graph is split-normal. These operations preserve the triangle hypergraph and therefore preserve ν and τ; splitting can only increase the number of vertices.

A triangle-isolated K_4 edge block would, by Lemma 2.3, leave a graph G' with

$$\nu(G') = 3, \qquad \tau(G') \geq 7 > 2\nu(G'),$$

contradicting Parker's theorem (2). Similarly, a nonempty crown of size h would, by Lemma 2.2, leave

$$\nu(G') = 4 - h \leq 3, \qquad \tau(G') - 2\nu(G') = \tau(G) - 8 + h > 0,$$

again contradicting (2).

Choose a maximal packing and saturate it under (E1)–(E3). All hypotheses of Lemma 3.1 now hold, so the split graph has at most $3|S| \leq 3\nu(G) = 12$ vertices. The original triangle support has no more vertices than its split graph, proving the proposition. □

4 Marked four-packing types

Let $P = \{T_1, T_2, T_3, T_4\}$ be four edge-disjoint triangles. Any two members of P share at most one vertex. For every vertex lying in at least two members, record the non-singleton subset of [4] consisting of the triangles that contain it. No pair $\{i, j\}$ occurs in two recorded subsets, and every index occurs at most three times. Private vertices fill the remaining incidences so that every T_i has size three.

Conversely, any family of non-singleton subsets of [4] with these two properties defines a linear four-edge, 3-uniform hypergraph, hence a marked four-packing type. There are at most six shared vertices because they consume at least two of the twelve incidences. We quotient only by the 24 permutations of the marked triangles. We do not quotient the ambient graph.

For an ambient order n, include every marked type whose union uses at most n vertices; the remaining vertices are outside the marked packing. The 16 canonical incidence families are listed in Appendix A. Counting each family at every ambient order at least its support order gives Table 1.

Table 1: Marked support types used at each ambient order.

order n	6	7	8	9	10	11	12	total
types	1	3	6	11	14	15	16	66

Orders below six are impossible: four edge-disjoint triangles use twelve distinct graph edges, while K_5 has only ten. Every graph with $\nu(G) = 4$ contains a maximum marked four-packing.

Relabelling its packing blocks and vertices places it in one of the 66 order-type scopes above. Thus the marked scope is complete for every triangle-supported graph of order at most twelve.

5 The SAT encoding

Fix an order n and one marked type. For every possible graph edge $e \in \binom{[n]}{2}$, introduce a variable x_e. For every possible triangle $T \in \binom{[n]}{3}$, introduce a variable y_T. The base formula contains:

$$y_T \longrightarrow x_e \qquad (e \in \tbinom{T}{2}), \tag{4}$$

$$\bigwedge_{e \in \binom{T}{2}} x_e \longrightarrow y_T, \tag{5}$$

$$x_e \longrightarrow \bigvee_{T \supset e} y_T, \tag{6}$$

$$\bigvee_{T \ni v} y_T \qquad (v \in [n]), \tag{7}$$

and a unit clause y_T for each of the four marked triangles. Equations (4)–(5) make the y-variables the complete literal triangle closure of the graph selected by the x-variables. Equations (6)–(7) make the graph triangle-supported.

The incremental search uses only two kinds of learned semantic clause.

Lemma 5.1 (packing cut)**.** *If a decoded model contains five edge-disjoint triangles $T_1, \dots, T_5$, then*

$$\neg y_{T_1} \vee \cdots \vee \neg y_{T_5}$$

is valid for every graph with $\nu \le 4$.

Lemma 5.2 (cover cut)**.** *If a decoded model has a triangle transversal C with $|C| \le 8$, then*

$$\bigvee_{T\colon E(T) \cap C = \varnothing} y_T$$

is valid for every graph with $\tau \ge 9$.

Proof. The packing clause merely says that the five displayed triangles cannot all exist. For the cover clause, a graph with no triangle disjoint from C has $C \cap E(G)$ as a transversal of size at most eight, even if some edges of C disappear in a later assignment. □

The algorithm solves the base formula, reconstructs the literal graph and its complete triangle set, and searches by exact recursion first for a five-packing and then for a transversal of size at most eight. It appends the corresponding valid clause and repeats; no additional graph-specific clause is used. When the final formula is unsatisfiable, every triangle-supported graph in that marked scope has either a five-packing or an eight-edge transversal.

6 Certificate verification

Each case is encoded directly as a SAT instance over the possible graph edges, with semantic clauses obtained from exact packing and covering calculations. The Lean development [1, 8] reconstructs every serialized witness, validates all 331,430 semantic clauses, checks their exact agreement with

the final DIMACS formulas, and discharges each of the 66 formulas by a kernel-checked LRAT refutation. An independent executable verifier performs the same witness reconstruction and formula comparison, and the original DRAT traces are checked separately with `drat-trim`.

A separate coverage checker reconstructs every expected marked type at each order and verifies that the resulting 66 cases occur exactly once.

Table 2 records the complete computation. Here "models" means decoded assignments that produced a semantic cut; it does not denote isomorphism classes.

Table 2: Certified finite computation.

order	cases	models	five-packing cuts	cover cuts	result
6	1	4	0	4	UNSAT
7	3	220	96	124	UNSAT
8	6	2,955	1,865	1,090	UNSAT
9	11	18,660	12,963	5,697	UNSAT
10	14	50,781	36,413	14,368	UNSAT
11	15	95,176	67,324	27,852	UNSAT
12	16	163,634	113,286	50,348	UNSAT
total	66	331,430	231,947	99,483	UNSAT

All 66 DRAT proofs were accepted by `drat-trim`; together they contain 570,595,138 bytes. Their 572 binary LRAT segments occupy 250.98 MiB. Each formula, proof, computation record, and verification program is identified by its SHA-256 digest.

The formal proof covers the graph-theoretic definitions, Parker's packing-rank-three bound, the twelve-vertex reduction, the classification of marked four-packings, the graph-to-CNF interpretation, and the final case dispatch. The SAT solver, witness-generation scripts, and DRAT-to-LRAT converter are not trusted: their outputs are checked by the formal development. The trusted base consists of the Lean kernel, the imported Mathlib definitions and proof infrastructure, and the checked source and certificate data.

Proposition 6.1. *Every triangle-supported graph G on at most twelve vertices with $\nu(G) = 4$ has $\tau(G) \leq 8$.*

Proof. Mark a maximum four-packing. By Section 4, the marked graph belongs to one of the 66 scopes. The corresponding final formula is DRAT-refuted. Lemmas 5.1 and 5.2 show that every graph with $\nu \leq 4$ and $\tau \geq 9$ satisfies every learned clause. Hence no such graph exists in the marked scope. □

7 Proof of the main theorem

Proof of Theorem 1.1. If $\nu(G) \leq 3$, the result is Parker's theorem (2). Suppose $\nu(G) = 4$ and, for a contradiction, $\tau(G) \geq 9$. Put $H = \mathrm{supp}_{\triangle}(G)$. Proposition 1.2 and (3) give $|V(H)| \leq 12$, $\nu(H) = 4$, and $\tau(H) = \tau(G)$. Applying Proposition 6.1 to the triangle-supported graph H gives $\tau(G) = \tau(H) \leq 8$, a contradiction. □

8 Concluding remarks

Theorem 1.1 extends the exact inequality from packing number three to packing number four. The reduction in Proposition 1.2 is specific to this bounded case and does not provide an induction on the packing number. Extending the method beyond rank four will therefore require an additional structural principle.

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A Canonical marked-packing incidence families

For compactness, write 13 for the shared vertex belonging to marked triangles T_1 and T_3, and similarly write 234 or 1234 for a vertex shared by three or four marked triangles. The complete canonical list is:

support order	non-singleton incidence subsets
6	$12, 13, 23, 14, 24, 34$
7	$12, 13, 23, 14, 24$
7	$12, 13, 14, 234$
8	$12, 13, 23, 14$
8	$12, 13, 24, 34$
8	$12, 13, 234$
9	$12, 13, 23$
9	$12, 13, 14$
9	$12, 13, 24$
9	$12, 134$
9	1234
10	$12, 13$
10	$12, 34$
10	123
11	12
12	none

Every row satisfies the pair-occupancy and degree conditions in Section 4. Conversely, recursively add a non-singleton subset of [4] only when it repeats no pair and raises no block degree above three; then quotient the resulting family by the 24 permutations of [4]. This produces exactly the rows above. Private vertices fill each block to degree three, giving the displayed support order. Taking all rows of support order at most n yields the cumulative counts $1, 3, 6, 11, 14, 15, 16$ for $n = 6, \ldots, 12$ and therefore 66 order-type scopes.

References

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